

SiTA: Exploiting Sparsity in Tensor-Product for Accelerating Quantum Readout Error Mitigation

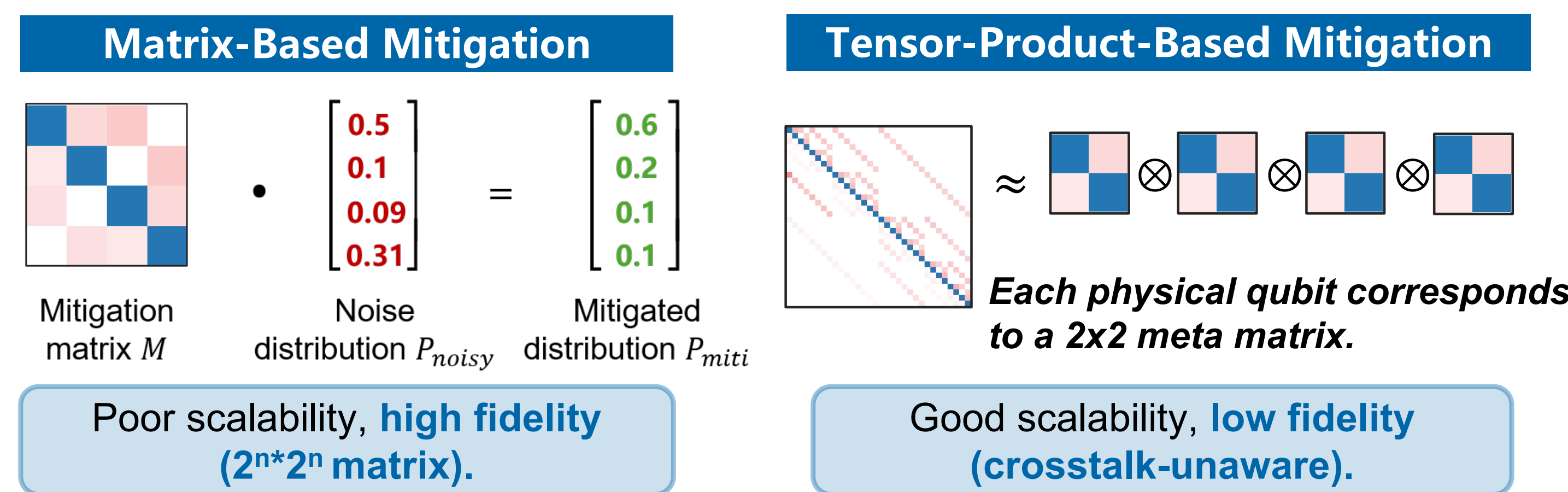
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Quantum Readout Error Mitigation

Quantum readout aims to convert quantum information into classical signals; however, this process is susceptible to noise-induced errors, thereby compromising information fidelity.

Existing mitigation methods:

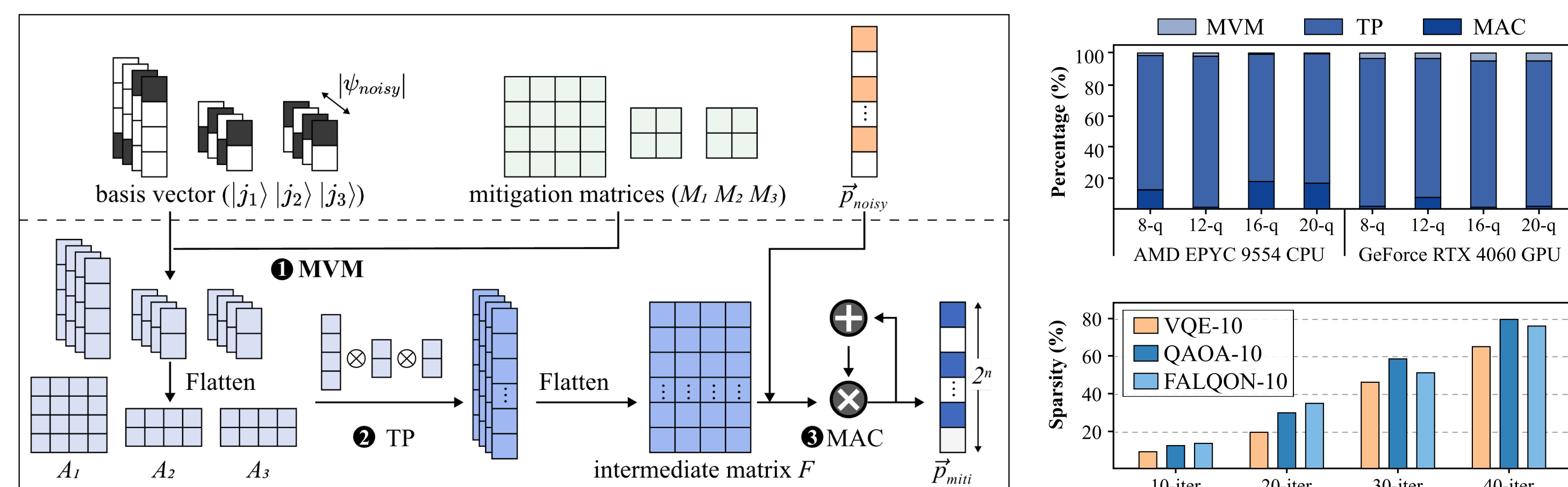


Iterative refinement can improve the fidelity of tensor-product-based mitigation.

Motivation

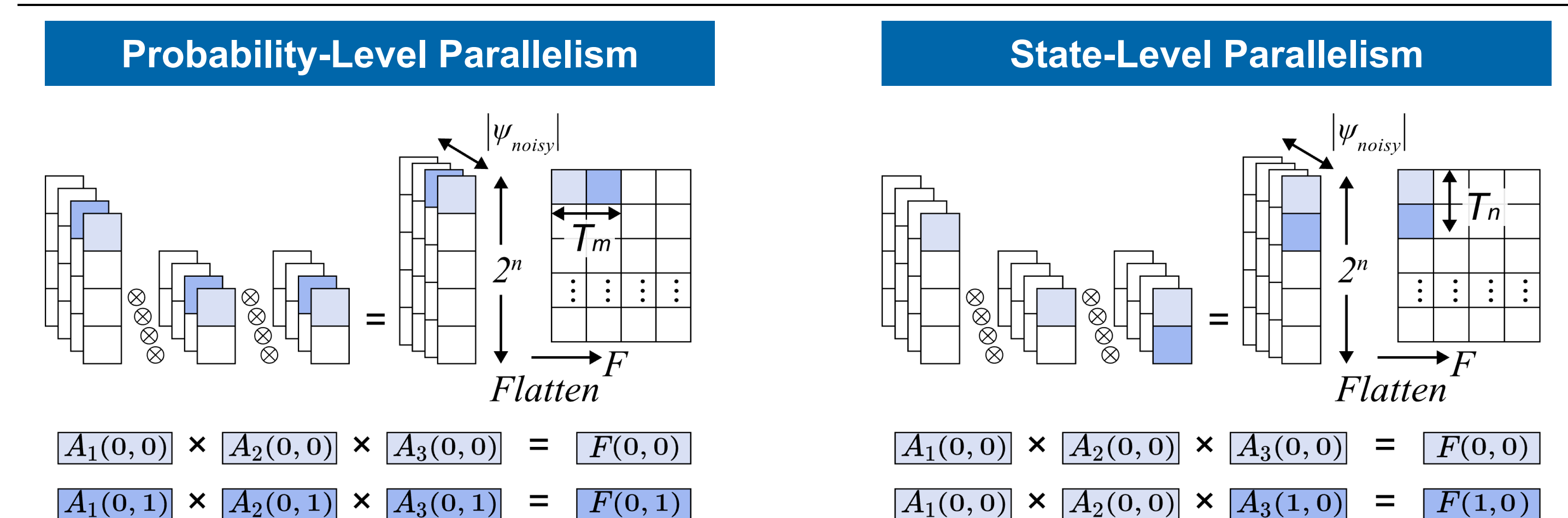
Goal: improve the efficiency of iterative tensor-product-based mitigation.

Tensor-product-based mitigation consists of three core operators: matrix-vector multiplication (MVM), tensor product (TP), and multiply-accumulate (MAC).



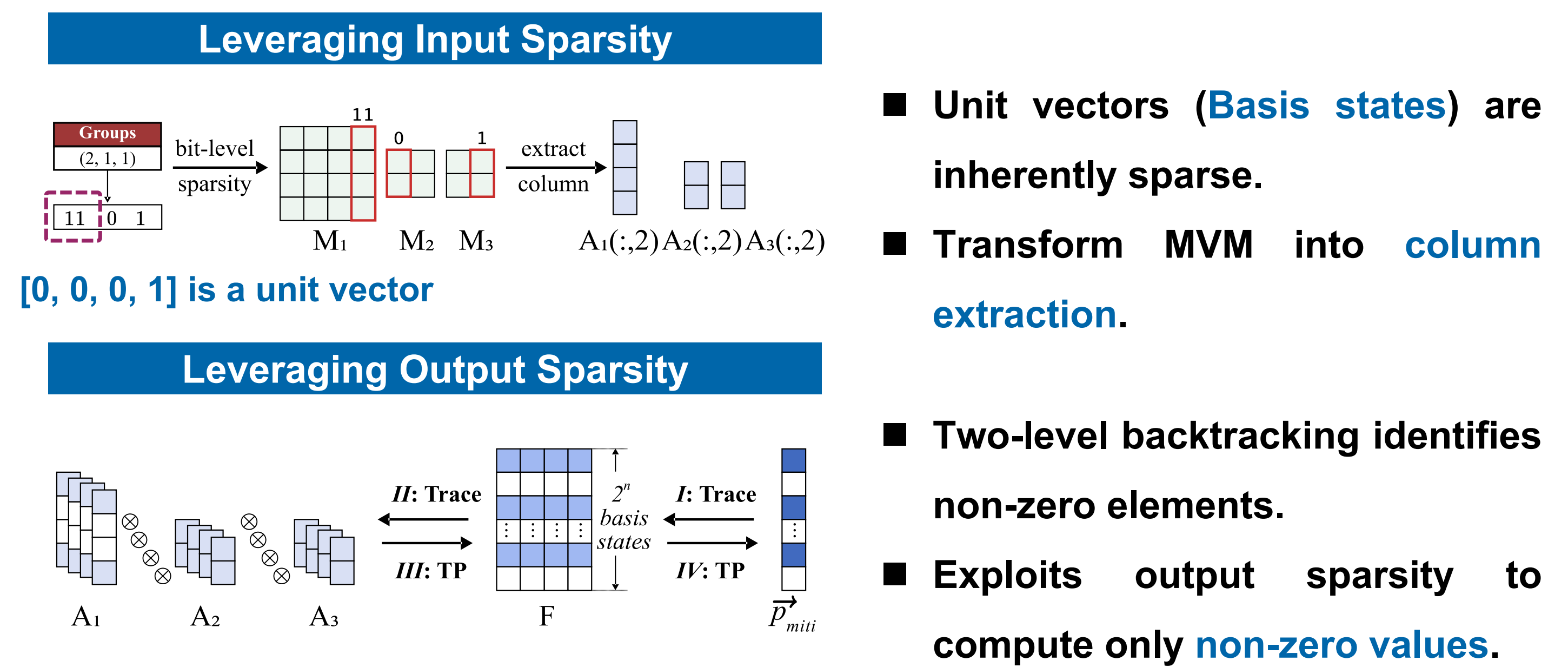
The TP operator is the bottleneck. Both the TP and MAC operators contain substantial redundant computations due to sparsity.

Parallelization



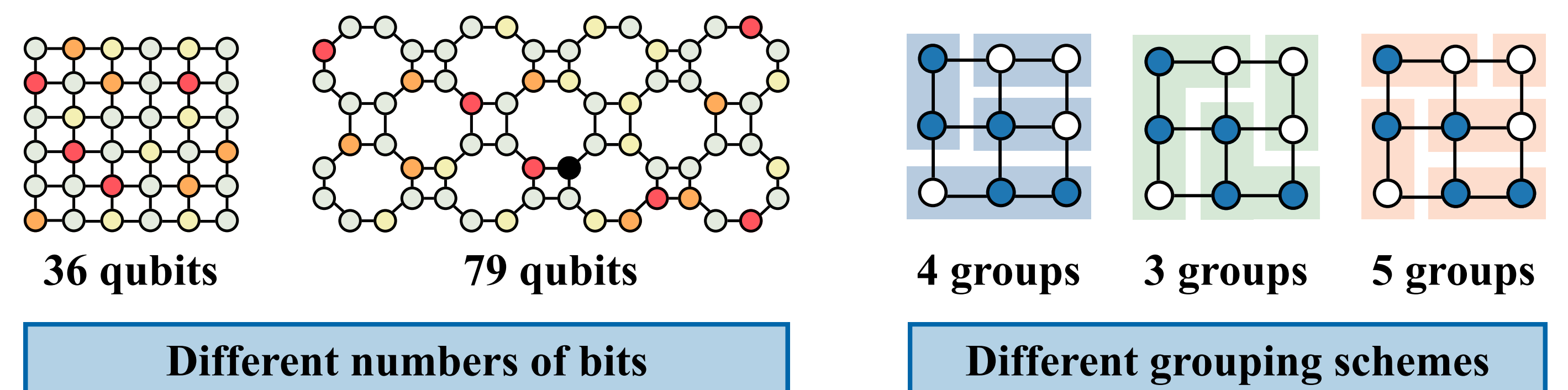
Combining both parallelization strategies accelerates TP computation to one tile per cycle.

Leveraging Sparsity

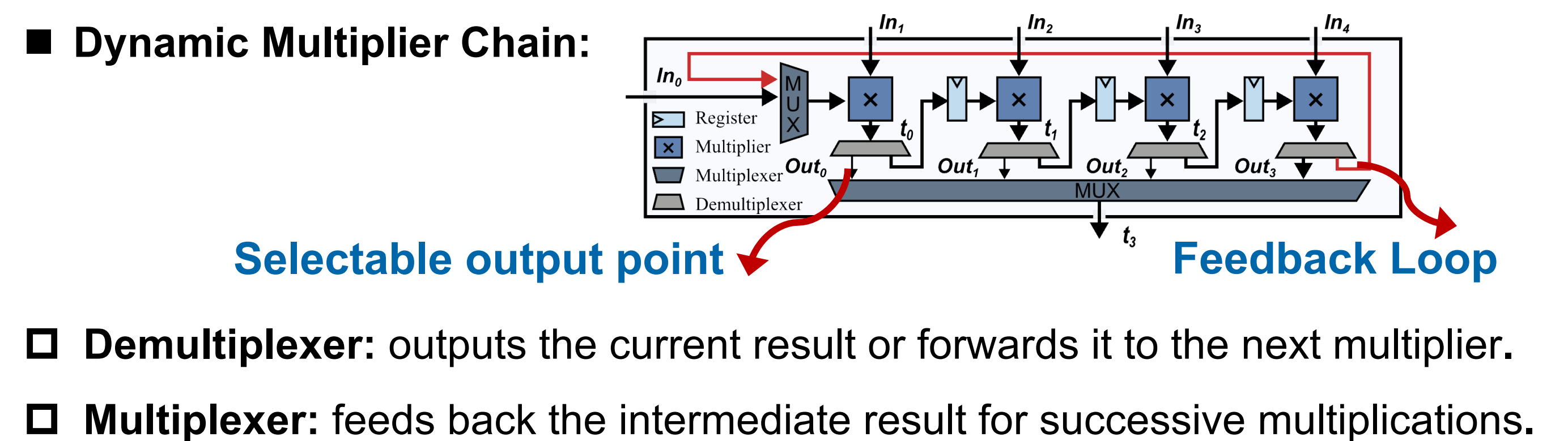


Dynamic Multiplier Chain

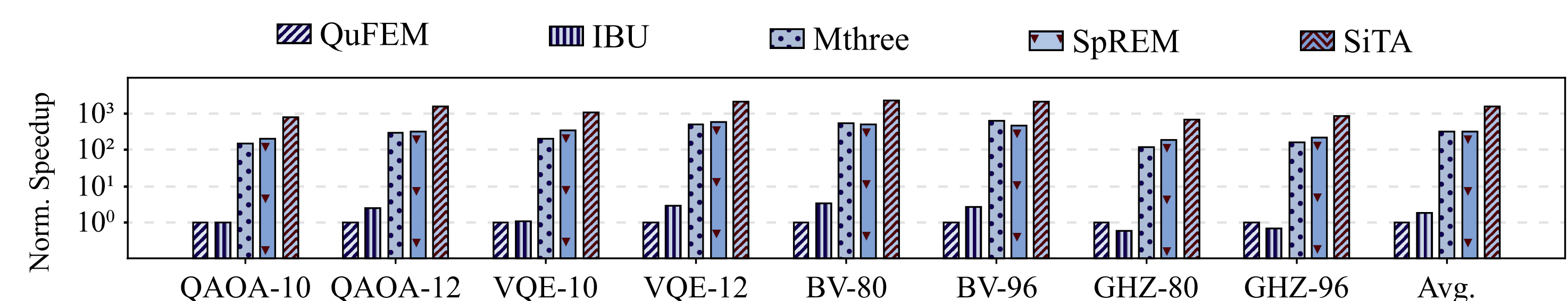
A flexible processing unit is needed for chained multiplications of varying lengths.



To address this, we design a flexible dynamic multiplier chain.



Evaluation



SiTA achieves an average speedup of $5.0\times$, $4.9\times$, $844\times$, and $1604\times$ over SpREM [1], Mthree [2], IBU [3], and QuFEM [4], respectively.

Reference

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- N. et al., "Scalable mitigation of measurement errors on quantum computers," *PRX Quantum*, 2021.
- S. et al., "Realizing topologically ordered states on a quantum processor," *Science*, 2021.
- T. et al., "Quferm: Fast and accurate quantum readout calibration using the finite element method," in *ASPLOS*, 2024.